
Intrinsic Motivation for Exploration in MiniGrid: RIDE and Object-Centric Curiosity*

Eric Xia

Department of Computer Science
Stanford University
Stanford, CA 94305
ruichu.xia@stanford.edu

Qi Wu

Department of Computer Science
Stanford University
Stanford, CA 94305
q1wu@stanford.edu

Abstract

We study intrinsic motivation for exploration in sparse-reward reinforcement learning using MiniGrid environments [1]. We implement and compare two approaches: novelty-based curiosity that rewards approaching and interacting with novel objects, and RIDE (Rewarding Impact-Driven Exploration), which uses state representation changes as intrinsic rewards [2]. We address a known issue in RIDE where agents favor turning actions in first-person view environments by introducing action-specific reward scaling. Experiments on MiniGrid navigation tasks show that both methods improve exploration and task performance over a baseline PPO-LSTM agent. Our results suggest that combining object-aware novelty signals with state-change-based exploration can improve sample efficiency in sparse-reward environments.

1 Introduction

1.1 Background

In reinforcement learning, an agent interacts with an environment modeled as a Markov decision process (MDP), selecting actions to maximize the expected discounted sum of rewards. In many interesting tasks, including those studied in this work, extrinsic rewards are extremely sparse: the agent receives non-zero feedback only upon reaching a distant goal state. In such settings, naive exploration strategies (e.g., ϵ -greedy or entropy regularization) are often insufficient, as the probability of randomly stumbling upon a rewarding trajectory is extremely low, leading to prohibitively high sample complexity.

Intrinsic motivation addresses this challenge by augmenting the environment reward with an internally generated signal that encourages exploration of “interesting” states. Prior work has instantiated intrinsic rewards using state visitation counts and pseudo-counts, prediction error in a learned dynamics model, or the error of a predictor network trained against a fixed random target [3–6]. More recently, impact-based methods such as RIDE reward actions that cause large changes in a learned state representation, combined with episodic normalization to prevent repeatedly exploiting a few high-impact transitions [2]. These approaches have been particularly effective in sparse-reward environments where most states are seen only once.

1.2 Problem Statement

MiniGrid provides a family of lightweight, configurable grid-world environments designed specifically for research on exploration, generalization, and memory [1]. In this work, we focus on a set

*Code available at <https://github.com/Ruichu-Xia/minigrid-with-curiosity>.

of canonical MiniGrid benchmarks—DoorKey and MultiRoom—to study how different intrinsic motivation mechanisms interact with a recurrent PPO agent in these challenging settings:

Procedurally generated environments: The use of Procedural Generation ensures that agents encounter novel environment layouts, object placements, and required action sequences in every episode. This requires generalization over memorization and demands the agent to develop strategies to explore the environment within each new episode for certain task (e.g. four rooms). The procedural generated environment is a major source of state uncertainty.

Partial observability: The agent is restricted to a local view window and cannot see objects beyond solid objects such as walls, further adding to state uncertainty. Consequently, the agent cannot determine its global state or the location of critical objects (like a distant key or the goal) from a single observation. This means the agent must retain memory about where it has visited to build a coherent representation of the full, unobserved state of the environment in order to solve the task successfully.

Sparse extrinsic rewards: The most distinctive challenge of MiniGrid is the extreme sparsity of extrinsic feedback, where non-zero reward is provided only upon successful completion. Difficult tasks in MiniGrid require the agent to either perform a long, precise sequence of interactions such as in door key or sufficiently explore the entire procedurally generated environment such as in multi-rooms before receiving any reward signal.

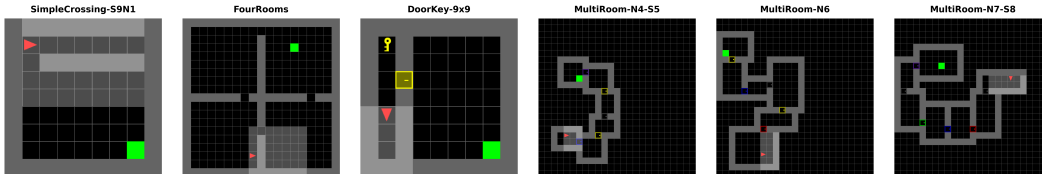


Figure 1: MiniGrid Environments. SimpleCrossing-S9N1 (Reach the green goal square on the opposite side of the room), FourRooms (Navigate through four rooms connected by doorways to find the goal), DoorKey-9x9 (Pick up a key to unlock a door and reach the goal square), and MultiRoom-N4-S5, N6, N7-S8 (Explore complex, procedurally generated mazes of increasing size to locate the final goal).

2 Related Work

Intrinsic motivation and curiosity. Many methods address sparse rewards by adding intrinsic bonuses that encourage exploration. Count-based and pseudo-count approaches derive bonuses from state visitation statistics or density-model uncertainty, extending tabular exploration to high-dimensional states [3, 4]. Curiosity-based methods instead use prediction error in learned models: the Intrinsic Curiosity Module (ICM) rewards forward-model error in a feature space trained via an inverse dynamics objective [5], while Random Network Distillation (RND) measures novelty as the error of a predictor network trained to match a fixed random target [6]. These techniques are effective on Atari and 3D control benchmarks but are mostly evaluated in fixed or slowly changing environments.

Impact-based exploration and RIDE. RIDE introduces an intrinsic reward proportional to the change in a learned state embedding, normalized by episodic visitation counts to avoid repeatedly exploiting a few high-impact states [2]. It achieves strong performance on procedurally generated MiniGrid tasks compared to prior curiosity and count-based approaches, but exhibits a turning bias in first-person settings where small rotations produce large visual changes. We build directly on RIDE: using modern MiniGrid environments [1], we analyze its behavior on long-horizon navigation tasks and introduce an action-dependent scaling that down-weights turning while preserving impact-driven exploration.

Intrinsic motivation in MiniGrid. MiniGrid is a standard benchmark for exploration and generalization, providing procedurally generated, partially observable grid-world tasks with sparse rewards [1]. Beyond RIDE, several works design intrinsic objectives specialized to MiniGrid, such as skill-based curiosity that rewards progress on learned sub-goals [7]. These methods typically operate in abstract skill or latent spaces. In contrast, our novelty-based curiosity module uses MiniGrid’s symbolic

object representation to define object-centric approach and interaction bonuses, combined with a simple episodic count-based term. Together with our RIDE variant, this places our work at the intersection of state-change-based, count-based, and object-centric intrinsic motivation for exploration in procedurally generated environments.

3 Method

3.1 PPO with Recurrent Memory

To handle partial observability in MiniGrid, we use a recurrent actor-critic architecture optimized via Proximal Policy Optimization (PPO) [8]. Since the agent receives only a 7×7 egocentric view, we incorporate a Long Short-Term Memory (LSTM) network [9] to retain interaction history.

Observations are processed by three convolutional layers and fed into the LSTM, whose hidden state drives separate actor and critic heads. We optimize the network using the standard PPO clipped objective to ensure stable policy updates.

This PPO-LSTM setup serves as the backbone upon which we build the intrinsic motivation modules described in the following sections.

3.2 RIDE with PPO

To address the extreme reward sparsity in MultiRoom tasks, we augment the PPO agent with RIDE [1, 2]. RIDE introduces an intrinsic reward that measures how much the agent’s actions impact the environment in a learned representation space. Let $\phi(\cdot)$ be a learned state embedding of observations. The intrinsic reward at time t is defined as

$$r_{\text{int}}(t) = \|\phi(s_{t+1}) - \phi(s_t)\|_2, \quad (1)$$

and the agent optimizes a combined reward signal

$$r_{\text{tot}}(t) = r_{\text{ext}}(t) + \beta r_{\text{int}}(t), \quad (2)$$

where r_{ext} is the environment reward and $\beta > 0$ controls the strength of intrinsic motivation. The PPO objective $\mathcal{L}_{\text{PPO}}$ is applied unchanged, but with advantages A_t computed from returns under r_{tot} .

Following the original RIDE formulation, we normalize intrinsic rewards by an episodic state visitation count to discourage the agent from repeatedly exploiting a small set of high-impact transitions. Let $N_{\text{ep}}(s)$ denote the number of times a state s has been visited in the current episode. The normalized intrinsic reward is

$$\tilde{r}_{\text{int}}(t) = \frac{\|\phi(s_{t+1}) - \phi(s_t)\|_2}{\sqrt{N_{\text{ep}}(s_{t+1})}}, \quad (3)$$

so that revisiting the same state yields diminishing intrinsic returns. In our implementation, we additionally apply a small extra penalty when $N_{\text{ep}}(s_{t+1})$ exceeds a threshold, further reducing oscillation between a few states.

The embedding $\phi(\cdot)$ is trained jointly with the agent using self-supervised dynamics objectives. We use a forward dynamics model f_{fwd} to predict the next embedding from the current embedding and action, and an inverse dynamics model f_{inv} to predict the action from consecutive embeddings:

$$\mathcal{L}_{\text{fwd}} = \|\phi(s_{t+1}) - f_{\text{fwd}}(\phi(s_t), a_t)\|_2, \quad (4)$$

$$\mathcal{L}_{\text{inv}} = \text{CrossEntropy}(a_t, f_{\text{inv}}(\phi(s_t), \phi(s_{t+1}))). \quad (5)$$

The overall loss combines PPO and dynamics terms,

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{PPO}} + \lambda_{\text{fwd}} \mathcal{L}_{\text{fwd}} + \lambda_{\text{inv}} \mathcal{L}_{\text{inv}}, \quad (6)$$

with coefficients λ_{fwd} and λ_{inv} controlling the strength of the auxiliary tasks. When computing intrinsic rewards, we explicitly detach $\phi(s_t)$ and $\phi(s_{t+1})$ from the computation graph to prevent the policy from directly shaping the embedding in order to artificially inflate r_{int} .

Finally, *MiniGrid*’s partially observed environment setting makes turning actions particularly prone to exploitation: small rotations can produce large visual changes and thus large intrinsic rewards [2]. To mitigate this, we introduce an action-dependent scaling of intrinsic rewards:

$$\hat{r}_{\text{int}}(t) = \alpha(a_t) \tilde{r}_{\text{int}}(t), \quad (7)$$

where $\alpha(a_t) = \tau$ for turning actions and $\alpha(a_t) = 1$ for all other actions, with $0 < \tau \leq 1$ a turn-action penalty hyperparameter. This makes turning relatively less attractive from the perspective of intrinsic motivation, reducing pathological spinning while preserving the exploratory benefits of RIDE.

3.3 Novelty-based Curiosity

Taking inspiration from the human tendency to approach and interact with visually distinctive objects, the Novelty-based Curiosity module is designed to reward the agent for approaching and interacting with distinctive elements that stand out from the observation, such as keys, doors, and boxes.

To define distinctive elements, the Symbolic Distinctiveness Detector identifies actionable objects (e.g., keys, doors, balls) by excluding static background elements (walls, floors) from the symbolic observation. The Uninteracted Object Tracker maintains a lifetime interaction count $N_{\text{interact}}(o)$ for every object signature (type and color), which determines each object’s novelty value $\mathcal{N}(o)$ decaying exponentially to ensure diminishing returns for repeated interactions:

$$\mathcal{N}(o) = e^{-0.01 \cdot N_{\text{interact}}(o)} \quad (8)$$

The core NoveltyApproachReward module combines two scaled components. The Approach Reward is calculated by comparing distinctive object positions across consecutive timesteps, rewarding movement towards a novel object. The Interaction Reward explicitly rewards successful interactions with novel objects that do not lead to a negative reward - such as picking up or toggling an object, or reaching the goal. The object-based novelty reward, r_{novelty} , is the scaled sum:

$$r_{\text{novelty}} = \beta_{\text{approach}} \cdot r_{\text{approach}} + \beta_{\text{interaction}} \cdot r_{\text{interaction}} \quad (9)$$

The specific reward components are calculated as follows:

Approach Reward ($\mathcal{R}_{\text{approach}}$): rewards the reduction of distance to any novel object o and is calculated by summing the contribution of all objects for which the distance change is positive ($\mathbf{1}[\Delta d_o > 0]$).

$$r_{\text{approach}}(t) = \sum_{o \in \text{objects}} \Delta d_o \cdot \mathcal{N}(o) \cdot \mathbf{1}[\Delta d_o > 0] \quad (10)$$

Interaction Reward ($\mathcal{R}_{\text{interaction}}$): provides a bonus for every successful interaction with object o (pickup, toggle, or goal-reach), scaled by the object’s novelty prior to the interaction.

$$r_{\text{interaction}}(t) = \sum_{o \in \text{interacted}} \mathcal{N}^{\text{pre}}(o) \quad (11)$$

In large state space tasks (e.g., MultiRoom), the agent requires a mechanism to encourage broad spatial coverage. The Count-Based Exploration Reward provides an episodic state visitation bonus to drive the agent into unexplored areas. This module uses an episodic visit count $N_{\text{ep}}(s)$ that tracks the number of times each hashed observation has been visited within the current episode, resetting at the start of every new attempt. The reward r_{count} is calculated as:

$$r_{\text{count}}(s_{t+1}) = \beta_{\text{count}} \cdot \frac{1}{\sqrt{N_{\text{ep}}(s_{t+1})}} \quad (12)$$

The total reward $\mathcal{R}_{\text{total}}$ used for training the agent for this approach is the weighted sum of all components:

$$r_{\text{total}} = \beta_{\text{ext}} \cdot r_{\text{env}} + \beta_{\text{int}} \cdot r_{\text{novelty}} + \beta_{\text{count}} \cdot r_{\text{count}}$$

4 Experiments and Results

We evaluate our approaches across 6 MiniGrid tasks shown in Figures 1. We quantify success by whether the agents are able to solve the task (achieving a stable high extrinsic reward) and their convergence speed, measured by the number of training frames.

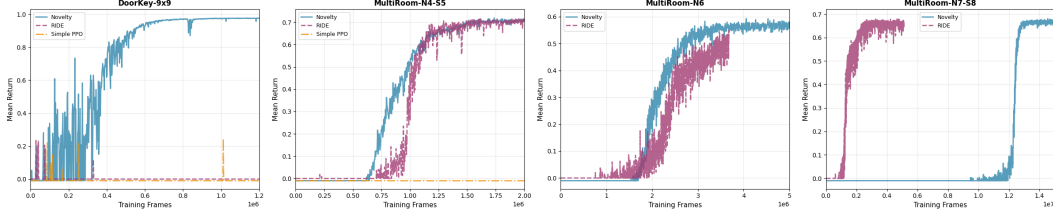


Figure 2: Intrinsic Motivation Comparison. Novelty excels on interaction-heavy tasks (DoorKey-9x9), RIDE shows faster initial convergence on spatial tasks (MultiRoom), and Simple PPO fails universally.

4.1 Simple PPO

The simple PPO-LSTM baseline agent successfully solves two easy tasks, SimpleCrossing and FourRooms, achieving stable high returns 3a–3b. Experiments confirm that the LSTM layer is necessary for FourRooms since an agent without memory cannot track previously visited rooms under partial observability. However, on the sparse-reward DoorKey and MultiRoomN4 tasks, the baseline fails to learn a successful policy even after 2×10^6 frames. Specifically, returns remain near zero on MultiRoomN4, and while the agent encounters sporadic rewards on DoorKey, it fails to consolidate these into a stable policy 4a–4b.

4.2 RIDE

We evaluate PPO+RIDE on four sparse-reward MiniGrid tasks: DoorKey-9x9-v0, MultiRoom-N4-S5-v0, MultiRoom-N6-v0, and MultiRoom-N7-S8-v0. For all environments we keep the PPO-LSTM and RIDE hyperparameters fixed and train for 2–5M environment frames; training curves are shown in Figures 5a–6b.

On DoorKey-9x9-v0, RIDE yields only occasional successful episodes: average return stays near zero with a best value of ≈ 0.24 , and episode lengths remain close to the horizon. Losses are stable but policy entropy stays relatively high, indicating that impact-based intrinsic reward alone is not sufficient to reliably discover the precise key–door sequence.

On the MultiRoom mazes, RIDE is much more effective. For N4-S5, performance improves rapidly after about 8×10^5 frames and stabilizes around 0.7 average return, while episode length drops from ~ 120 to ~ 40 steps and entropy decays toward zero. As the number of rooms increases (N6, N7-S8), learning becomes slower but follows the same pattern: after a long exploration phase, returns rise to ≈ 0.6 – 0.68 and episode length falls to roughly 50–60 steps, with well-behaved actor/critic losses. Overall, RIDE substantially improves exploration and task completion in long-horizon corridor-style mazes, but struggles on structured object-interaction tasks like DoorKey-9x9.

4.3 Novelty-based Curiosity

We evaluate the Novelty-based Curiosity agent on the same set of sparse-reward MiniGrid tasks as the RIDE framework. We maintained fixed PPO-LSTM hyperparameters for all environments, training for 2–5M frames for most tasks, except for MultiRoom-N7-S8, which required over 10M frames to stabilize 7a–8b.

On DoorKey-9x9-v0 task, the novelty agent demonstrates rapid and reliable convergence, successfully solving the task where the RIDE baseline failed. The distinctive object novelty reward quickly enables the agent to learn to approach and interact with the key and the door. By learning that picking up the key and toggling the door are good actions, the agent is more likely to subsequently stumble across the goal, producing a significant extrinsic reward that quickly dominates the intrinsic signal. Learning grows significantly after approximately 3×10^5 frames, at which time the agent reliably reaches the goal within each batch of trajectories and successfully completes the task.

For MultiRoom mazes, the novelty agent remains fairly effective primarily due to its ability to encourage opening the doors, which are treated as distinctive objects. With only the novelty reward, the agent solves N4-S5 in little over 1×10^6 frames. The N6 environment proved more challenging

due to the expansion of the state space, taking over 3×10^6 frames to solve. For the most challenging large-scale task, N7-S8, the added Episodic Count-Based Exploration Reward becomes critical. This bonus encourages the agent to explore all unique states within an episode beyond simple door opening, making it more likely to discover the goal state. Nevertheless, the task complexity demanded extended training, with the agent solving the environment at 1.25×10^7 frames.

Overall, the novelty-based approach excels at tasks requiring explicit object interaction and successfully solves the most complex MultiRoom maze by leveraging the episodic count bonus for comprehensive spatial coverage.

5 Discussion

Both RIDE and the Novelty-based Curiosity module aim to mitigate the sparse reward problem by providing the agent with intrinsic, intermediate learning signals. However, they diverge in their definition of such signals: RIDE incentivizes exploration through significant changes in a learned state representation, whereas the Novelty module explicitly rewards interaction with distinct objects. Our experiments highlight a fundamental trade-off between these impact-driven and object-centric exploration strategies.

The failure of RIDE on DoorKey and its success on MultiRoom suggests that impact-driven rewards are highly effective for navigation but insufficient for precise object manipulation. In MultiRoom, moving between rooms causes significant changes in the agent’s egocentric view, generating large intrinsic rewards. However, in DoorKey, crucial actions such as picking up a key or toggling a locked door result in minimal visual changes to the state embedding. Consequently, the RIDE agent ignores these low-impact but necessary transitions.

In contrast, the Novelty-based approach succeeds in DoorKey because it leverages semantic distinctiveness rather than visual change. By assigning high intrinsic value to the first interaction with a key or door, the module creates an implicit sequence (find key $\rightarrow$ open door $\rightarrow$ reach goal) that guides the agent regardless of how small the visual change is.

The results on MultiRoom-N7-S8 reveal the limits of object-based curiosity. While object interactions effectively guide the agent through doors, they do not inherently encourage the traversal of the large empty spaces between distinct elements. This limitation required the addition of the Count-Based Exploration Reward to ensure comprehensive spatial coverage.

6 Conclusion and Future Work

In this work, we studied exploration in sparse-reward MiniGrid environments by comparing impact-driven (RIDE) and novelty-based intrinsic motivation. We showed that while the PPO-LSTM baseline fails in complex settings, adding intrinsic rewards significantly helps learning. Specifically, our results shows RIDE is better for navigation tasks because it rewards actions that cause significant changes in the state representation (like entering a new room), whereas the Novelty module performs better on object interaction tasks because it rewards interacting with distinct objects. We conclude that a more general hybrid approach is the path forward to create robust agents that can generalize across more diverse types of sparse-reward problems.

6.1 Limitations

A main limitation is that our method is sensitive to the balance between the scale of intrinsic and extrinsic rewards. If the intrinsic reward is too high, the agent might learn to repeatedly trigger the bonus (like toggling a door) rather than solve the task. This requires careful tuning for each specific environment.

Additionally, our Novelty module currently relies on the hard-coded symbolic object detection provided by MiniGrid. To be applied to general environments, this component must be replaced with a more general perception model that can identify distinct objects directly from raw pixel data.

6.2 Future work

Future work should focus on making the intrinsic module more robust and general, for instance by replacing the symbolic detector with a learned perception model to allow operation in any visual-based environment. We also aim to make the method less sensitive to reward scaling, potentially through adaptive normalization techniques. Finally, it would be interesting to explore if impact-driven and novelty-driven approaches can be combined, leveraging their complementary strengths to aid learning in complex environments that require both navigation and interaction.

Contributions

Eric implemented the PPO-LSTM baseline algorithm and training infrastructure, configured the MiniGrid environment setup, and developed a novelty-based curiosity approach for sparse-reward exploration in the DoorKey and MultiRoom task families. Qi implemented the RIDE intrinsic reward module, addressed a known issue in RIDE where agents favor turning actions in first-person-view environments by introducing action-specific reward scaling, and carried out training and evaluation on the DoorKey and MultiRoom task families.

References

- [1] Maxime Chevalier-Boisvert, Bolun Dai, Mark Towers, Rodrigo de Lazcano, Lucas Willems, Salem Lahlou, Suman Pal, Pablo Samuel Castro, and Jordan Terry. Minigrid & miniworld: Modular & customizable reinforcement learning environments for goal-oriented tasks. *CoRR*, abs/2306.13831, 2023.
- [2] Roberta Raileanu and Tim Rocktäschel. Ride: Rewarding impact-driven exploration for procedurally-generated environments, 2020. URL <https://arxiv.org/abs/2002.12292>.
- [3] Marc G. Bellemare, Sriram Srinivasan, Georg Ostrovski, Tom Schaul, David Saxton, and Rémi Munos. Unifying count-based exploration and intrinsic motivation. In *Advances in Neural Information Processing Systems*, volume 29, 2016. URL <https://arxiv.org/abs/1606.01868>.
- [4] Georg Ostrovski, Marc G. Bellemare, Aaron van den Oord, and Rémi Munos. Count-based exploration with neural density models. In *Proceedings of the 34th International Conference on Machine Learning*, 2017. URL <https://arxiv.org/abs/1703.01310>.
- [5] Deepak Pathak, Pulkit Agrawal, Alexei A. Efros, and Trevor Darrell. Curiosity-driven exploration by self-supervised prediction. In *Proceedings of the 34th International Conference on Machine Learning*, 2017. URL <https://arxiv.org/abs/1705.05363>.
- [6] Yuri Burda, Harrison Edwards, Amos Storkey, and Oleg Klimov. Exploration by random network distillation. In *International Conference on Learning Representations*, 2019. URL <https://arxiv.org/abs/1810.12894>.
- [7] Nicolas Bougie and Ryutaro Ichise. Skill-based curiosity for intrinsically motivated reinforcement learning. *Machine Learning*, 109(2):493–512, 2020. doi: 10.1007/s10994-019-05845-8.
- [8] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *CoRR*, abs/1707.06347, 2017. URL <https://arxiv.org/abs/1707.06347>.
- [9] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8): 1735–1780, 1997. URL <https://doi.org/10.1162/neco.1997.9.8.1735>.

A Appendix

Simple PPO Training results

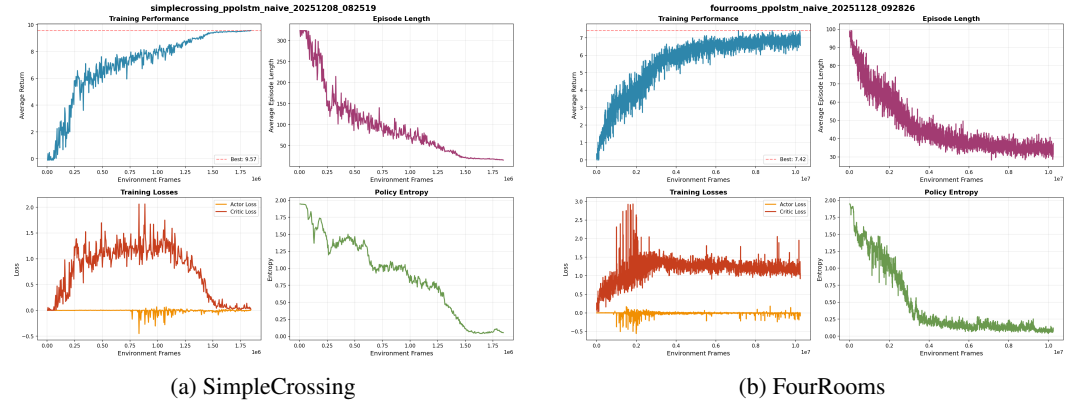


Figure 3: Simple PPO training curves on SimpleCrossing and FourRooms.

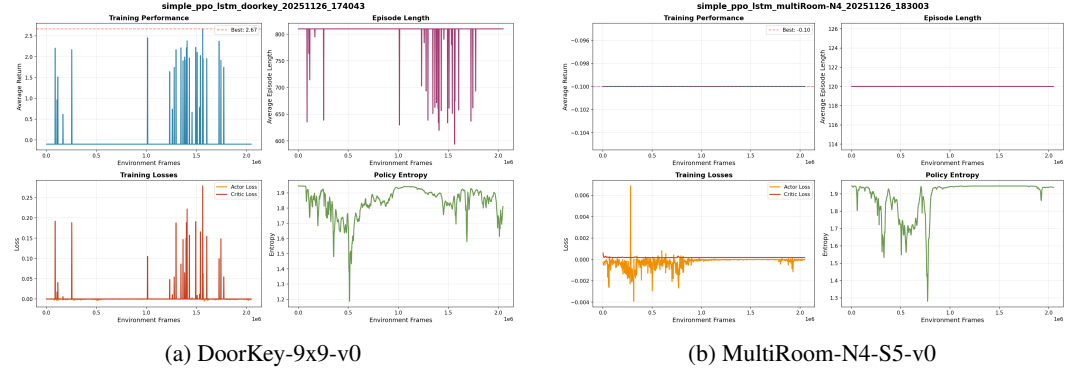


Figure 4: Simple PPO training curves on DoorKey and MultiRoom-N4-S5.

Ride Training results

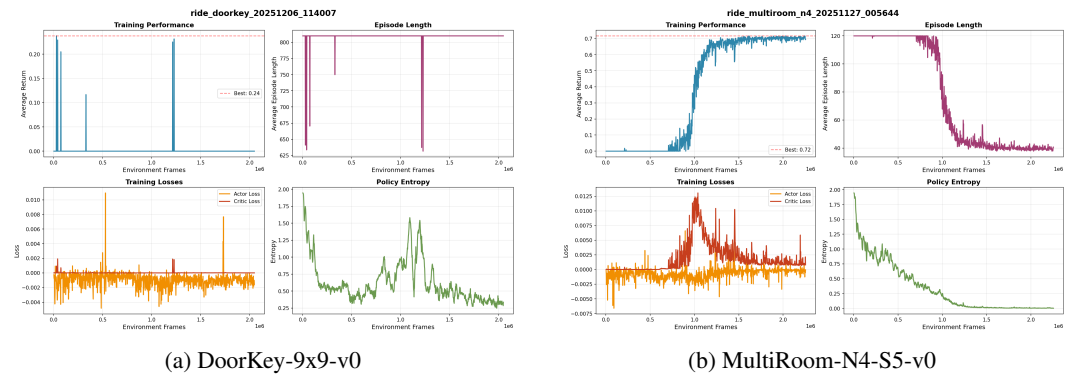
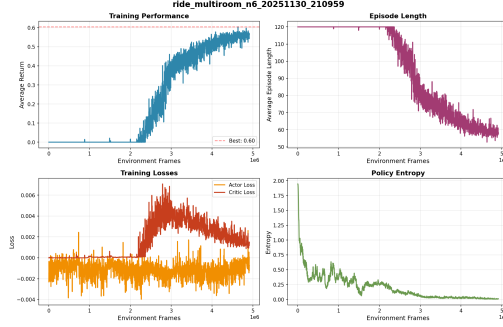
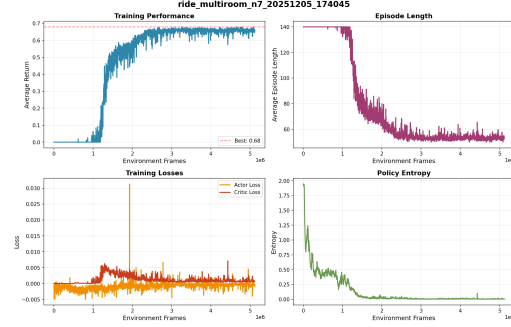


Figure 5: RIDE training curves on DoorKey and MultiRoom-N4-S5.



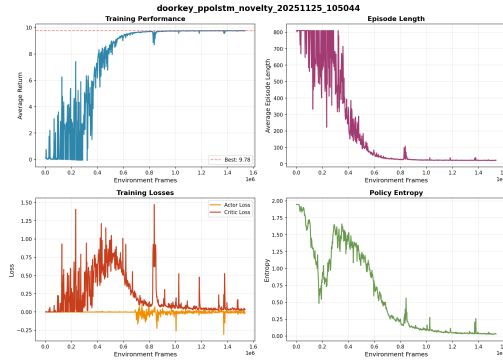
(a) MultiRoom-N6-v0



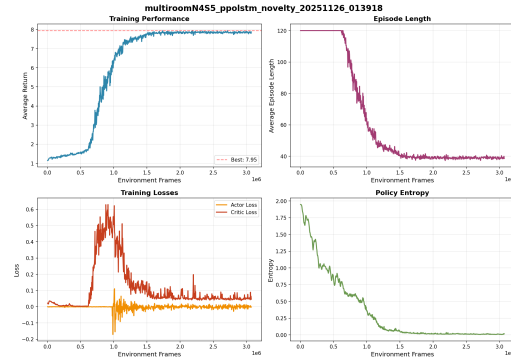
(b) MultiRoom-N7-S8-v0

Figure 6: RIDE training curves on larger MultiRoom mazes.

Novelty-based Curiosity Training results

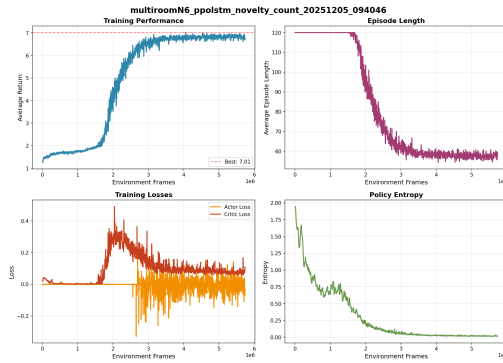


(a) DoorKey-9x9-v0

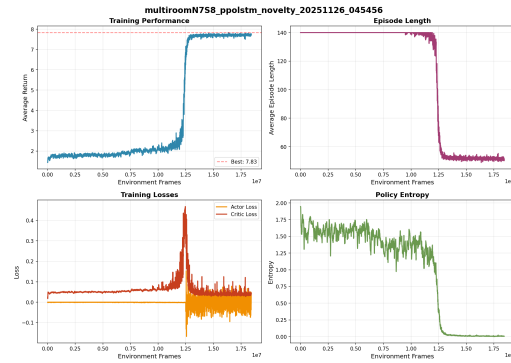


(b) MultiRoom-N4-S5-v0

Figure 7: Novelty-based curiosity training curves on DoorKey and MultiRoom-N4-S5.



(a) MultiRoom-N6-v0



(b) MultiRoom-N7-S8-v0

Figure 8: Novelty-based curiosity training curves on larger MultiRoom mazes.

Training Configuration

Table 1: RIDE configuration.

Parameter	Value
Experiment name	ride_{task_name}
Environment name	{env_name}
Device	CPU or CUDA
Method	RIDE
PPO iterations	5000
Steps per iteration	2048
Learning rate	3×10^{-4}
Discount factor γ	0.99
GAE λ	0.95
PPO epochs	4
Minibatch size	64
Clip range ϵ	0.2
Value loss coef.	0.5
Entropy coef.	0.002
Max grad norm	2
Clip value loss	False
LSTM hidden size	256
Max sequence length	128
Intrinsic reward coef.	0.01
Forward loss coef.	0.1
Inverse loss coef.	0.1
Intrinsic norm.	True
Turn action penalty	0.1
Save interval	every 100 iters
Print interval	every 10 iters
Plot interval	every 100 iters

Table 2: PPO-LSTM novelty configuration.

Parameter	Value
Experiment name	{task_name}
Environment name	{env_name}
Device	CPU or CUDA
Method	PPO-LSTM + novelty
PPO iterations	10000
Steps per iteration	4096
Learning rate	3×10^{-4}
Discount factor γ	0.99
GAE λ	0.95
PPO epochs	4
Minibatch size	64
Clip range ϵ	0.2
Value loss coef.	0.5
Entropy coef.	0.01
Max grad norm	2
Clip value loss	False
LSTM hidden size	256
Use curiosity	True
Curiosity approach scale	0.02
Curiosity interaction scale	1
Extrinsic reward scale	10.0
Intrinsic reward scale	0.1
Save interval	every 100 iters
Print interval	every 1 iter
Plot interval	every 25 iters